HIRA AHMED

AI Engineer | Backend & Automation | QA Lead

Karachi, Pakistan · Remote · hira229922@gmail.com · linkedin.com/in/hira-ahmed-4068402a7 · github.com/hira299 · hira299.github.io

PROFESSIONAL SUMMARY

AI Engineer specializing in production AI systems, automation, backend engineering, and software quality. Experienced in building LLM pipelines, AI agents, n8n automations, API integrations, and AWS-backed systems, with a strong focus on reliability, failure handling, and verification. Built and operated production systems processing 28,000+ records, improved pipeline throughput by 5.6×, and reduced monthly AWS infrastructure costs by 34%. Also leads QA across four client products of multi-tenant B2B software, documenting 1,500+ defects while testing end-to-end business workflows, APIs, authorization, data integrity, and security-sensitive edge cases. Lead author and developer of Sentinel-Mesh, a neuro-symbolic research framework combining LLM-generated Terraform remediation with Z3 SMT formal verification.

PROFESSIONAL EXPERIENCE

TechPotion.ai, Remote · Malaysia-based *Jan 2025 – Present*

Quality Assurance Lead *Sep 2026 – Present*

QA Lead across four concurrent client contracts, owning defect discovery, API testing, authorization, multi-tenant isolation, and business-logic validation.

Tekruiter · Lead QA Contract

- Surfaced 60+ defects across 20+ modules of a multi-tenant recruitment platform (CRM, ATS, timesheets, invoicing) by leading QA and a second tester.
- Protected data confidentiality by exposing leaks of candidate, pay, and margin data to unauthorized users and other agencies' jobs on the candidate portal.
- Safeguarded billing accuracy by catching a $7.95M miscalculated billable total, pre-start-date hour logging, and a database error blocking invoice creation.

PharmaConnect · QA Contract

- Uncovered 110+ defects across 15 modules by leading QA through 20 end-to-end journeys and 7 user roles, with API-level reproduction evidence.
- Exposed critical security flaws, including authentication bypass, privilege escalation, cross-tenant account tampering, and session tokens and OTPs leaked in audit logs.
- Prevented financial corruption by catching overpayments, doubled invoice quantities, and expired-discount pricing, then verifying fixes through retesting.

ToolPotion · QA Contract

- Documented 1,300+ defects across 10 QA rounds for a 28K+ AI tools directory, including 170+ critical ranking, categorization, and data-integrity failures.
- Protected users by flagging phishing, malware, and gambling-redirect listings and 190+ dead or fake tools, then verified 300+ fixes through retesting.

AI Cloud Academy · QA Contract

- Documented 80+ defects across 5 QA rounds and 37 courses, including critical blockers: mobile users unable to navigate or progress, lessons locking on rapid clicks, and level certificates not issued.
- Eliminated a gamification exploit (XP over-awarded on rapid clicks) and verified 40+ fixes through regression testing, plus 68 video-content recommendations.

Backend & Automation Engineer *Jan 2025 – Sep 2026*

- Increased pipeline throughput by 5.6× across 28,000+ records by re-architecting ETL into a fault-tolerant parallel system with 50 concurrent workers, dynamic rate limiting, and circuit breakers.
- Reduced monthly AWS infrastructure costs by 34% through query optimisation and execution-flow refactoring across Lambda, RDS, EC2, S3, API Gateway, and EventBridge.
- Achieved ~95% data-extraction success rate by building Python scrapers (BeautifulSoup, Playwright, Selenium) across 10 platforms with 20+ extraction methods and a URL-healing service.
- Enabled 100% schema-valid LLM output for categorisation, FAQ generation, and 9-language translation by enforcing JSON-mode on a 7-stage status-gated enrichment pipeline via OpenRouter.
- Shipped 12+ production automation workflows using n8n and Make.com alongside FastAPI, PostgreSQL, Docker, Next.js, and LLM APIs.

Sadiq.ai, Karachi, Pakistan *Jun 2024 – Aug 2024*

QA Intern

- Improved dev-team bug-query efficiency by ~40% by building a RAG-based internal chatbot (Gemini API, Python, Streamlit) to surface defects from tracking data in real time.
- Reduced release-blocking defects by documenting 150+ edge-case bugs across web and mobile through manual and automated Flutter testing (unit, widget, golden tests).

RESEARCH

Sentinel-Mesh: Formally Verified Remediation of Cloud Misconfigurations · Lead Author & Developer · 2026

Neuro-symbolic framework combining LLM-generated Terraform remediation with Z3 SMT formal verification. Evaluated on CloudFix-Bench (105 AWS Terraform cases): achieved 83.81% remediation rate (88/105), 0.0% security regression, and 29 formal proof certificates. Preprint on Research Square (DOI: 10.21203/rs.3.rs-10674271/v1); under review at **IEEE Transactions on Cloud Computing** (Manuscript ID TCC-2026-08-0693). Artifact archived on Zenodo (DOI: 10.5281/zenodo.20975067).

SELECTED PROJECTS

Autonomous Cloud Compliance & AI Auditing Engine

LangGraph · CRAG · LlamaIndex · Supabase pgvector · FastAPI · n8n · AWS

- Cut questionnaire turnaround by ~90% by engineering an autonomous engine that auto-answers 250+ question enterprise reviews (CAIQ, SOC 2, SIG) from live AWS infrastructure evidence.
- Eliminated LLM hallucinations (100% citation enforcement) via a 6-node LangGraph CRAG state machine with Llama-3.1-8B relevance scoring and an automated Critic Agent verifying real AWS ARNs.
- Achieved 0.95 semantic relevance score at $0 embedding cost using PyMuPDF + FastEmbed all-MiniLM-L6-v2 stored in Supabase vector(384); processed 2,000+ question workloads with zero timeout failures.

Autonomous AI Email Agent

n8n · Gmail API · Groq · RAG · Google Sheets

- Eliminated manual screening overhead by building a stateful email agent with thread-level memory for automated candidate screening and technical Q&A, with explicit branching for malformed inputs.

Autonomous Competitor Intelligence & SEO Pipeline

n8n · ClickUp · Gemini · PostgreSQL

- Automated competitor research, SEO gap analysis, and brief generation by building a PostgreSQL-backed pipeline with dead-letter queue patterns ensuring 100% data integrity on concurrent updates.

TECHNICAL SKILLS

AI & LLM Systems	LangGraph, LangChain, LlamaIndex, RAG/CRAG, LLM pipelines, structured outputs, Groq, Gemini Flash, GPT-4o, Claude 3.5, OpenRouter, prompt engineering, pgvector, FastEmbed
Automation & Orchestration	n8n, Make.com, async state machines, concurrency locks, rate limiting, dead-letter queues, exponential backoff, idempotency, replay patterns
Backend & APIs	Python (advanced), FastAPI, Next.js 15, REST APIs, webhooks, OAuth, TypeScript, JavaScript, SQL
Cloud & Infrastructure	AWS (Lambda, RDS, EC2, S3, API Gateway, EventBridge, SQS, CloudWatch, IAM, VPC, CloudFormation), Terraform, Docker Compose, CI/CD, Linux/Bash
Databases	PostgreSQL (JSONB, GIN indexing, pgvector), Supabase, DynamoDB, NoSQL
QA & Reliability	QA team leadership, multi-tenant B2B QA, API testing, RBAC/authorization, business-logic testing, data-integrity testing, defect triage and severity classification, exploratory testing, regression testing, browser/network inspection
Security & Formal Verification	Z3 SMT solving, cloud security invariants, Terraform security, IAM policy analysis, security testing

EDUCATION & CREDENTIALS

B.S. Cybersecurity

Sir Syed University of Engineering and Technology (SSUET), Pakistan
Top 10 Finalist (Blue Team), Ignite National Hackathon (200+ teams)

Credentials & Open Source

ISC2 Certified in Cybersecurity (CC)
Verified Peer Reviewer, IEEE Access (5 reviews, Web of Science / ORCID)
n8n Production Resilience Patterns github.com/hira299/n8n-production-resilience-patterns
Enterprise Multi-Tenant QA Matrix github.com/hira299/enterprise-multitenant-qa-matrix